

Analysis.pdf Task 1: DNA/Protein Sequence Analysis

Tool name: NCBI BLAST. Identifying homologous proteins to **Human Lysozyme (P61626)** using BLAST.

Introduction: **BLAST** (Basic Local Alignment Search Tool) is one of the most widely used bioinformatics tools. The primary purpose of BLAST is to identify sequence similarities, detect potential species relatives, determine alignment scores, and locate conserved regions across organisms. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance.

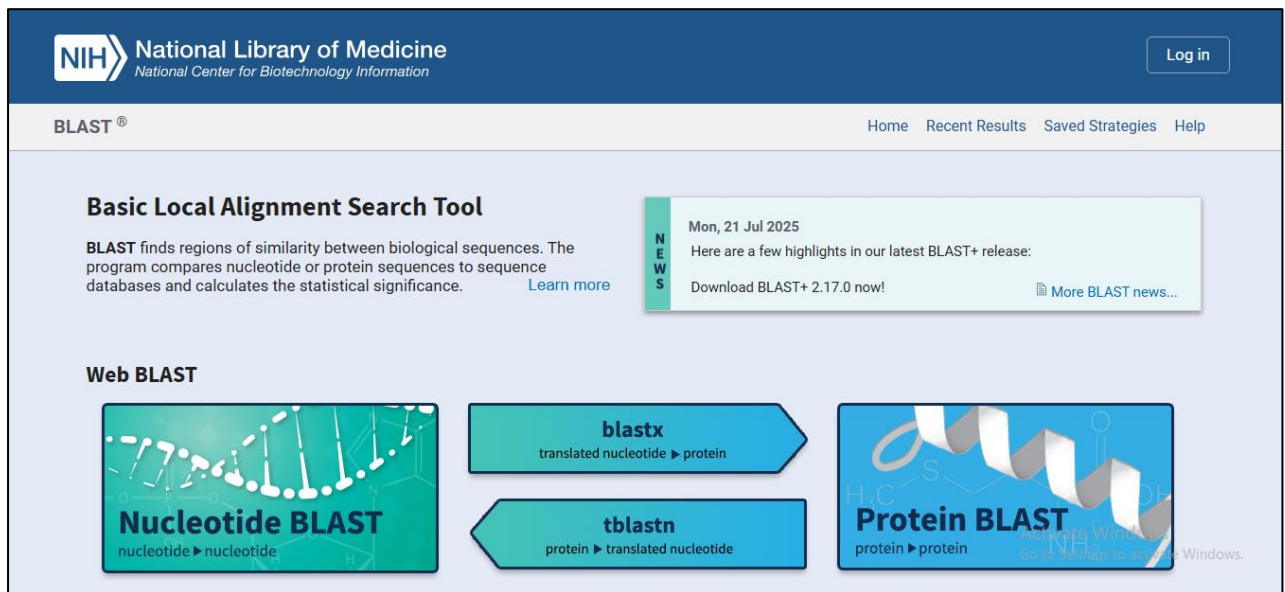


Fig 1: NCBI BLAST page (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>).

The image shows the NCBI BLAST search interface. At the top, there's a navigation bar with tabs for 'blastn', 'blastp', 'blastx', 'tblastn', and 'tblastx'. The 'blastp' tab is selected. Below this, the 'Standard Protein BLAST' section is visible. It includes a 'Enter Query Sequence' section with a text area containing a FASTA sequence of Human Lysozyme (P61626). There's also a 'Query subrange' section with 'From' and 'To' input fields. Below this, there's a 'Choose Search Set' section with a 'Database' dropdown set to 'ClusteredNR (nr cluster seq)' and an 'Organism' input field. The 'Program Selection' section has a radio button selected for 'blastp (protein-protein BLAST)'. At the bottom, there's a 'BLAST' button and a checkbox for 'Show results in a new window'.

Fig 2: FASTA sequence of Human Lysozyme (P61626).

Clusters		Graphic Summary		Alignments		Taxonomy																
Clusters producing significant alignments						Download		Select columns		Show	100											
☐ select all 0 clusters selected													GenPept		Graphics		Distance tree of results		Multiple alignment		MSA Viewer	
Cluster Composition		Cluster Ancestor		Cluster Representative Sequence		Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession										
Click the icon to see the cluster contents																						
☐	17 member(s), 7 organism(s)	apes	lysozyme C precursor [Gorilla gorilla gorilla]	301	301	100%	3e-103	99.32%	148	NP_001266591.1												
☐	6 member(s), 6 organism(s)	Old World monkeys	lysozyme C precursor [Papio anubis]	279	279	100%	2e-94	88.51%	148	NP_001106112.1												
☐	1 member(s), 1 organism(s)	other sequences	Mdc-hly hybrid protein, partial [synthetic construct]	274	274	88%	9e-92	100.00%	185	A1R95615.1												
☐	1 member(s), 1 organism(s)	Cape golden mole	PREDICTED: lysozyme C [Chrysoschloris asiatica]	272	272	100%	1e-91	86.49%	148	XP_006859607.1												
☐	1 member(s), 1 organism(s)	black-bearded tomb bat	lysozyme C [Taphozous melanopogon]	271	271	100%	3e-91	83.78%	148	AFU52840.1												
☐	170 member(s), 2 organism(s)	human	Chain A LYSOZYME [Homo sapiens]	269	269	88%	1e-90	99.23%	130	1GB7_A												
☐	7 member(s), 7 organism(s)	bats	c-type lysozyme [Leptonycteris yerbabuenae]	269	269	100%	2e-90	86.49%	148	QEQ43372.1												
☐	8 member(s), 7 organism(s)	primates	lysozyme C [Ptilocolobus tephrosceles]	268	268	100%	3e-90	82.43%	148	XP_023057414.1												
☐	1 member(s), 1 organism(s)	degu	lysozyme C [Octodon degus]	262	262	100%	2e-87	83.11%	148	XP_004632212.1												
☐	2 member(s), 1 organism(s)	Iberian mole	lysozyme C [Talpa occidentalis]	261	261	100%	3e-87	81.08%	159	XP_037369906.1												
☐	3 member(s), 2 organism(s)	guinea pigs and others	PREDICTED: lysozyme C-like [Chinchilla lanigera]	258	258	100%	6e-86	81.08%	148	XP_005400871.1												
☐	1 member(s), 1 organism(s)	Greater mouse-tailed bat	lysozyme C [Rhinopoma microphyllum]	257	257	100%	1e-85	80.41%	148	KAN3857697.1												

Fig 3: BLAST results displayed similar organisms, percentage identity, and query coverage.

Clusters

Graphic Summary

Alignments

Taxonomy

Alignment view

Pairwise

Restore defaults

Download

100 clusters selected

Download

GenPept

Graphics

Next

Previous

Descriptions

lysozyme C precursor [Gorilla gorilla gorilla]

Sequence ID: NP_001266591.1 Length: 148 Number of Matches: 1

Range 1: 1 to 148

GenPept

Graphics

Next Match

Previous Match

Score	Expect	Method	Identities	Positives	Gaps
301 bits(772)	3e-103	Compositional matrix adjust.	147/148(99%)	147/148(99%)	0/148(0%)
Query 1	MKALIVLGLVLLSVTVQGVFERCELARTLKRLGMDGYRGISLANMCLAKWESGYNTRA				60
Sbjct 1	MKALIVLGLVLLSVTVQGVFERCELARTLKRLGMDGYRGISLANMCLAKWESGYNTRA				60
Query 61	TNYNAGDRSTDYGIQINSRYMNDGKTPGAVNACHLSCSALLQDNIADAVACAKRVVRD				120
Sbjct 61	TNYNAGDRSTDYGIQINSRYMNDGKTPGAVNACHLSCSALLQDNIADAVACAKRVVRD				120
Query 121	PQIRAWVAWRNRCQNRDVRQYVQGC		148		
Sbjct 121	PQIRAWVAWRNRCQNRDVRQYVQGC		148		

Related Information

Gene - associated gene details

AlphaFold Structure - 3D structure displays

Genome Data Viewer - aligned genomic context

Fig 4: BLAST results displayed alignment score.

Analysis:

The BLAST search successfully generated a list of organisms with sequences similar to the query. The result showing **maximum query coverage and alignment score** possesses **Lysozyme C precursor (Gorilla gorilla gorilla)**; the **Mdc-hly hybrid protein** shows the highest percentage identity, which is **100%** against the Human Lysozyme C query, and a bit score of 305. These closely matching species are considered phylogenetically related, and the information obtained can be further applied to construct a phylogenetic tree and perform multiple sequence alignment.